ecosystem. The process of creating semiconductors spans several stages and requires a diverse array of inputs—from specialised gases and chemicals to sophisticated robotics and AI algorithms. However, MSMEs do not need to tackle all aspects. They can focus on specific niches where they have expertise or where costs are feasible.

For example, the first pillar, IC design, is perhaps the most accessible. Each design consists of several IPs that can be independently designed and developed by Indian startups. MSMEs could also develop specialised software for various semiconductor applications, such as automotive systems like battery management or charging infrastructures, where the value-added significantly exceeds the production cost. Even small components like analogue front ends or A/D converters present opportunities. These are critical to larger systems and often monopolised by a few large companies, allowing smaller players to enter with competitive pricing.

Moreover, the burgeoning demand in sectors like automotive, IoT, and industrial electronics in India provides a robust market for innovative semiconductor products. MSMEs can harness this demand by developing niche products or services, with strong potential for growth by offering competitive alternatives to the high-margin products of dominant players.

Thus, while the semiconductor sector is capital-intensive, strategic entry points exist for MSMEs to contribute meaningfully and profitably. This approach supports the national objective of strengthening semiconductor capabilities and positions these enterprises to capitalise on India's rapid economic growth and increasing technological needs.

Sanjay's Background

Sanjay graduated in electronics engineering from Delhi College of Engineering in 1996. He later completed an MBA at the Indian School of Business, Hyderabad, earning recognition on the Dean's List. After his engineering, he accepted an offer from a US-based startup. Within a year, he was transferred to San Jose, California, where he gained insights into the burgeoning field of microprocessors and microcontrollers. His initial projects involved 250-nanometre technology. By 1998, that startup was acquired by Motorola, where he helped establish their wireless design centre in India, later transitioning to their networking group to work on high-performance chips.

From 2005 to 2016, he created India's largest automotive R&D centre, which had over 500 US patents. After Motorola transitioned to NXP, he served as the Vice President of Engineering and Managing Director for NXP's India Technical Innovation Centres, overseeing 5000 engineers working on innovative projects in the industrial, IoT, networking and automotive sectors. He currently leads L&T's nascent semiconductor venture as Chief Development Officer and Global Head of Engineering at L&T Semiconductor Technologies (LTSCT). Prior to joining L&T Semiconductor, Sanjay spearheaded Minda Corporation's Advanced Technologies business vertical as President & CEO.

How can MSMEs identify their niche in the market, especially when facing budget constraints that limit their ROI compared to giant corporations like L&T or Tata?

Identifying the right niche for companies involves deep introspection and recognising the real strength areas with the potential for high ROI. I previously mentioned design, but there are multiple sectors within electronic manufacturing where MSMEs can excel. The challenge often lies in the awareness and the need for more knowledge about potential partners for creating synergies. For instance, L&T Semiconductors constantly seeks startups for acquisitions or partnerships in very specific niche areas of semiconductor design, but we do not find relevant matches. MSMEs must focus on solving real customer problems—this is often where many startups fail and end up developing Me-Too products. They either do not meet market needs or fail to align their offerings with customer issues. I advise MSMEs to stay as close to their customers as possible. Understanding customer needs sharpens focus on developing the right product architecture, increasing the chances of success by aligning with industry gaps.

Considering some entities' control over the raw material market, do you see a role for MSMEs in this sector?

Raw materials encompass a variety of elements depending on the context. For example, in electronics, it could mean supplying PCB boards and discrete components, which are crucial for maintaining supply chains. In my experience, many design projects are delayed due to unavailable components. MSMEs can establish themselves by offering optimal component solutions that big players could use to optimise their overall BOM cost or improve their supply chains. They could become key suppliers of high-quality, cost-effective components.

Furthermore, the definition of raw materials can extend beyond physical components. Talent can also be considered a critical raw material. MSMEs could distinguish themselves by supplying highly skilled professionals in specific fields. Another area is the chemical and gas supplies needed for semiconductor manufacturing. MSMEs could become distributors or even partners in joint ventures, potentially gaining a monopoly with strategic resource management.

The semiconductor industry is set to grow significantly, supported by national initiatives. MSMEs should prepare to scale their operations to align with industry growth timelines, like Tata's projected advancements by 2027. The key is to be proactive and ready when market demands increase, maximising their impact in the sector.